Jeffrey Blay

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Professional Summary: Applied Science Ph.D. student specializing in Geospatial Data Science, with 5+ years of experience in machine and deep learning for spatial analytics. Driven to develop advanced geospatial technologies that support evidence-based decisions, address complex societal challenges, and improve quality of life through data-informed solutions.

Certifications

- BM Professional Data Engineering Certificate — Coursera (In Progress, Expected May 2026)

Education

North Carolina A&T State University, Ph.D. Applied Science and Technology (August 2023 – Present)

- **Specialization:** Geospatial Data Science
- **Relevant Courses:** Machine Learning and Data Mining, Big Data Analytics, Advanced Geospatial Analysis, Neural networks, Statistical Methods, Multivariate Statistics for Engineers

Yale School of the Environment, Master of Environmental Science (August 2021 – May 2023)

- **Specialization:** Environmental Data science for Urban sustainability and resilience
- **Relevant Courses:** Advanced Remote Sensing, SDGs & Implementation, Statistics for Environmental Sciences, Data Exploration & Analysis, Environmental Data Science, Introductory Machine Learning

University Of Ghana, Bachelor of Arts Geography with Political Science (2015 - 2019)

- **Concentration:** Geography with Political Science minor
- **GPA:** 3.81/4.0

Professional Experience

Remote Sensing & GIS Lab, North Carolina A&T State University — **Greensboro, NC**
Graduate Research Assistant, NASA Flood Project | Part-Time — **September 2025 - Present**

- Implementing CNN (UNet, Attention-UNet) and transformer (Swin-UNet, SegFormer) models for urban flood-depth prediction in PyTorch.
- Designing a physics-informed deep learning framework to integrate hydrostatic constraints; running controlled experiments benchmarking physics-informed vs. standard deep learning models.
- Building end-to-end full-stack interactive dashboard deployment framework with FastAPI and interactive dashboards for stakeholder flood risk and vulnerability assessment.

NASA DEAP Institute, North Carolina A&T State University — **Greensboro, NC**
Data Science Research Intern, NASA Data Project | Part-Time — **Summer 2025**

- Trained and evaluated deep learning models (U-Net, Attention U-Net, U-Net++) using TensorFlow for urban flood-depth prediction, producing visualizations (ArcGIS Online) to inform flood-risk assessment.
- Developed and published ezprocess, an open-source Python library for geospatial data preprocessing (PyPI/GitHub), reduced data preparation time by 70% across ML/DL workflows.
- Managed codes, experiments, and library version control using Git and GitHub, enabling collaborative development across research teams.

Remote Sensing & GIS Lab, North Carolina A&T State University — **Greensboro, NC**
Graduate Research Assistant, NASA Data Project | Part-Time — **September 2024 – May 2025**

- Engineered geospatial ETL pipelines in ArcGIS and Python (rasterio, ArcPy) to ingest, preprocess, and integrate multi-source geospatial datasets for ML training.
- Led 3-person team to design, build, and publish a public benchmark dataset, managing data schema, QA, and documentation from end to end.
- Implemented U-Net and pix2pix cGAN workflows to infer flood depth from UAV imagery and LiDAR point clouds.

GEMS Institute, North Carolina A&T State University - Greensboro, North Carolina-USA.
Data Science Research Assistant, NSF Multi Modal Data Fusion Project | Full-Time — **June - August 2024**

- Processed 700k+ 1-m resolution geospatial pixels via NumPy and scikit-learn pipelines for large-scale flood prediction with machine learning models (Random Forest & XGboost).

- Collaborated with a 5-member team to develop and publish a multi-source geospatial dataset for flood inundation modeling; designed automated pipelines preprocess and integrate multi-sensor datasets using ArcPy in ArcGIS Pro.
- Completed training workshop in cloud-based data storage management and retrieval using Amazon S3.

Remote Sensing & GIS Lab, North Carolina A&T State University- Greensboro, North Carolina-USA.
Graduate Research Assistant, NOAA Flood Project | Part-Time **September 2023 – May 2024**

- Engineered web-scraping workflows with python (Beautiful Soup) to curate a custom image classification training datasets.
- Built a framework using PyTorch to train and evaluate image classification models (GoogleNet, ResNet18) for automated flood classification of crowdsourced images.
- Led a technical assessment of GAN-based approaches for flood mapping using remote sensing.

Urban Resources Initiative, Non-Profit Organization **New Haven, CT**
Community Group Manager, Summer Greenspace Community Initiative | Full-Time **Summer 2023**

- **Supervised** and **managed** 7 different community greenspace groups.
- **Coordinated** weekly project initiatives and strategies for fostering the green goals of each community group.
- **Engaged** with each community group to **design** and **visualize technical ideas** using **ArcGIS Pro and QGIS**.

Yale School Of The Environment **New Haven, CT**
Teaching Fellow: Real-world Environmental Data Science Course | Part-Time **January 2023 – May 2023**

- Held office meetings to assist 12 students with python programming assignments, projects, and general class issues.
- Assisted the Class to utilize Google Collaboratory, GitHub, and GitHub classroom for teaching.

Hixon Center For Urban Ecology **New Haven, CT**
Research Assistant, NASA Environmental Justice Project | Part-Time **September 2022 – May 2023**

- Utilized multi-spectral satellite imagery and geostatistical techniques to conduct census block-level analysis to identify UHI-vulnerable urban communities.
- Conducted a block level analysis of Surface temperature using ArcPy and ArcGIS Pro SQL query builder.
- Used R (sp, dplyr, ggplot2) for time-series statistical analysis and mapping of urban heat and tree cover gaps.

Tropical Research Institute, Yale University **New Haven, CT**
Research Fellow, TRI Research Fellowship | Part-Time **August 2022 – May 2023**

- Developed deep learning datasets and trained a UNet regression model for predicting building composition in metropolitan Ghana using satellite imagery.
- Leveraged Google Earth Engine to process VIIRS night-time light data for analyzing electricity distribution.
- Applied geostatistics to identify and map electricity infrastructure gaps in built-up areas.

Tropical Research Institute, Yale University **Ghana**
Field Geospatial Data Scientist, Summer Research | Full-Time **Summer 2022**

- Designed and deployed a **real-time data pipeline** using **ArcGIS Field Maps** and **Information Dashboards** to collect and analyze **300+** ground truth building data.
- Enabled **real-time data visualization** for data collection **improve data accuracy by 70%** for geospatial modeling.
- Applied **geo-statistics** to identify and map electricity infrastructure gaps in built-up areas.

Seto Lab, Urban Research Lab – Yale University **New Haven, CT**
Geospatial Research Assistant, Urban Africa Project | Part-Time **September 2021 - May 2022**

- Developed and optimized **Google Earth Engine workflows** for **large-scale satellite data** acquisition and preprocessing.
- Trained satellite image segmentation models in Python on a **high-performance Linux computing cluster**, implementing scalable batch processing pipelines.
- Built **custom R and Python scripts** for geospatial data processing, analysis, and visualizations.

Ghana Statistical Service, National Government Agency **Accra, Ghana**
GIS Data Officer, National Census Secretariat | Full-Time **September 2020 – June 2021**

- Performed spatial restructuring and quality assurance of Supervisory Area (SA) boundary maps, validating and correcting digitized census geospatial datasets.
- Collaborated with a 15-member Team to produce daily near real-time spatial coverage analytics to monitor enumerator field operations during national housing and population census activities.

Remote Sensing/GIS Lab, Department of Geography-University of Ghana **Accra, Ghana**
Teaching & Research Assistant | Full-Time **September 2019 – August 2020**

- Held Remote Sensing and GIS tutorial sessions for 150 undergraduate Remote Sensing/GIS students.
- Administered and Graded assignments and GIS group projects for the students.
- Assisted to undertake lab projects.

Awutu Senya East Municipal Assembly, City Authority **Kasoa-Accra, Ghana**
Intern: Physical Planning Department | Full-Time **Summer 2018**

- Inspected land use activities within the assembly with Planning Officers.
- Used GIS techniques and Aerial images to analyze and coordinate land use activities within the Municipal Area.
- Built attribute data for land title registration and permit approval.

Publications

Blay, J., Fawakherji, M., & Hashemi-Beni, L. (2024, September 5). *Flood impact risk mapping in settlement areas from a 3D perspective: A case study of hurricane matthew*. 2024 IEEE International Geoscience and Remote Sensing Symposium, Athens, Greece. https://doi.org/10.1109/IGARSS53475.2024.10640634.

Blay, J., Gebregziabher, Y., Jha, M. K., & Hashemi Beni, L. (2026). Inundation2Depth: A multi-source dataset for floodwater depth estimation in urban areas. *Data in Brief*, *64*, 112347. https://doi.org/10.1016/j.dib.2025.112347

Blay, J., & Hashemi-Beni, L. (2025a). Geospatial and Deep Learning Approaches for Modeling Floodwater Depth in Urbanized Areas. *Remote Sensing*, *18*(1), 60. https://doi.org/10.3390/rs18010060

Blay, J., & Hashemi-Beni, L. (2025b). Pixels to Insights: Deep Learning for Floodwater Depth Mapping in Settlement Areas. *IGARSS 2025 - 2025 IEEE International Geoscience and Remote Sensing Symposium*, 1140–1144. https://doi.org/10.1109/IGARSS55030.2025.11242913

Blay, J., & Hashemi-Beni, L. (2025c, April). *Advanced Geo-Data Analytics and AI for 3D Flood Mapping to Protect Built Assets*. ISPRS Geospatial Week 2025, Dubai,UAE.

Blay, J., Seto, K. C., & Chen, T.-H. K. (2024). *Dark Development: Satellite Analysis of Building Density and Electricity Provision in Ghana's Urban Areas*. SSRN. https://doi.org/10.2139/ssrn.5038955

Fawakherji, M., Anokye, M., **Blay, J.**, & Hashemi-Beni, L. (2025). Multi-Resolution Data Fusion for Resilient Flood Mapping. *IEEE Access*, *13*, 202275–202294. https://doi.org/10.1109/ACCESS.2025.3636732

Fawakherji, M., **Blay, J.**, Anokye, M., Hashemi-Beni, L., & Dorton, J. (2025). DeepFlood for Inundated Vegetation High-Resolution Dataset for Accurate Flood Mapping and Segmentation. *Scientific Data*, *12*(1), 271. https://doi.org/10.1038/s41597-025-04554-3

Owusu, A. B., Mensah, C. A., **Blay, J.**, & Fynn, I. E. M. (2023). Urban Growth and Land Surface Temperature Dynamics: Lessons from Ghana. *Theoretical and Empirical Research in Urban Management*, *18*(3).

Syum Gebre, T., Endale Ashebir, S., **Blay, J.**, Anokye, M., Pandey, V., & Hashemi-Beni, L. (2025). Real-Time Traffic Insights With Physics-Informed Neural Networks: Integrating the Aw-Rascle Model and LLMs. *IEEE Access*, *13*, 181246–181266. https://doi.org/10.1109/ACCESS.2025.3618282

Conferences & Presentation

2025 IGARS (International Geoscience and Remote Sensing Symposium) Conference **August 7, 2025**.

- **Presentation Topic:** Pixels to Insights: Deep learning for Floodwater Depth Mapping in Settlement Areas.
- **Audience:** International

2025 ISPRS Geospatial Week **April 10, 2025**.

- **Presentation Topic:** Advanced Geo-Data Analytics and AI for 3D Flood Mapping to Protect Built Assets

- **Audience:** International

2025 Annual ASPRS and Geo Week Conference **February 12, 2025**.

- **Presentation Topic:** 3D Floodwater Depth Analytics with Big Geo-Data and cGAN.
- **Audience:** International

2024 Annual AGU (American Geophysical Union) Conference **December 11, 2024**.

- **Presentation Topic:** Urban Flood Depth Analytics: Leveraging Generative Algorithms and Remote Sensing for 3D Flood Depth Simulation.
- **Audience:** International

2024 NCAUG (North Carolina ArcGIS Users Group) Conference **September 24, 2024**.

- **Presentation Topic:** Leveraging AI and big geo-data for 3D flood depth mapping to safeguard-built assets.
- **Audience:** United States of America

2024 IGARS (International Geoscience and Remote Sensing Symposium) Conference **July 08, 2024**.

- **Presentation Topic:** Flood Impact Risk Mapping in Settlement Areas from a 3D Perspective: A Case Study of Hurricane Matthew.
- **Audience:** International

2024 Annual ASPRS and Geo Week Conference **February 14, 2024**.

- **Presentation Topic:** A 3D perspective of analysing the effect of Hurricane Matthew in small communities: A case study of Grifton, NC.
- **Audience:** International

2023 Annual AGU (American Geophysical Union) Conference **December 14, 2023**.

- **Presentation Topic:** Multi-channel deep learning-based architecture for flood detection and mapping
- **Audience:** International

Tropical Resource Institute(TRI) Symposium **April 21, 2023**.

- **Presentation Topic:** Local Scale analysis of Urban Intensity and Electricity infrastructures in Ghana with multi-sensor satellite images and machine learning techniques; A case study of Metropolitan cities.
- **Audience:** TRI Directors and Fellows, and all interested persons .

39th Yale School of the Environment(YSE) Research Day **April 14, 2023**.

- **Presentation Topic:** Local Scale analysis of Urban Intensity and Electricity infrastructures in Ghana with multi-sensor satellite images and machine learning techniques; A case study of Metropolitan cities
- **Audience:** The Entire Yale Community and all interested persons.

Yale School of the Environment(YSE) Confluence Talk Series **March 24, 2023.**

- **Presentation Topic:** Local Scale analysis of Urban Intensity and Electricity infrastructures in Ghana with multi-sensor satellite images and machine learning techniques; A case study of Metropolitan cities
- **Audience:** Yale School of the Environment Community and all interested persons.

Honors & Awards

Lidar Leader Award **2024**

Best Poster presentation – International Conference, 2024 ASPRS & Geo Week, Denver-CO

Grants & Fellowships

Graduate Research Assistant Fellowship **2023 - 2026**

North Carolina A&T State University

Teaching Fellowship **2023**

Yale School of the Environment(YSE)

TRI Research Endowment Fellowship **2022**
Tropical Resource Institute, Yale University

YSE need based and Merit Scholarship (Master's Study) **2021 – 2023**
Yale School of the Environment (YSE)

Projects

Weather Alert Dashboard **Live** Python · FastAPI · JavaScript · Docker · CI/CD · REST API · GitHub Pages **2026**

- Built real-time weather monitoring platform integrating National Weather Service API; REST API with filtering, analytics, and risk-scoring across all 50 U.S. states.

Ezprocess - Open-Source Python Library Python · PyPI · GitHub · Geospatial ETL **2025**

- Designed, packaged, and published geospatial data preprocessing library to PyPI/GitHub; reduces preparation time by 70% .

Technical Skills

Programming Languages: Python, R, SQL, JavaScript
Geospatial & Remote Sensing: ArcGIS Suite, QGIS, Google Earth Engine, ArcPy, GDAL, ENVI, SNAP
ML/ DL Frameworks: PyTorch, TensorFlow, Scikit-Learn, Computer vision, Physics-Informed Neural Networks (PINNs)
Data Engineering & Processing: ETL Pipelines, Pandas, RDBMS, Web Scraping, Geospatial Data Processing
Visualization & Decision Support: Matplotlib, Seaborn, Dashboards, ArcGIS StoryMaps, Power BI, R Shiny (R), Tableau
Web, APIs & Applications: FastAPI, RESTful APIs, HTML/CSS, Streamlit
Cloud, DevOps & Infrastructure: AWS S3, Docker, CI/CD, Git, GitHub, HPC, GitHub Pages, Render
Databases: PostgreSQL, MySQL, IBM DB2

Leadership and Co-Curricular Activity

ASPRS Student Chapter – North Carolina A&T State University **Greensboro, NC-USA**
Secretary & Social Media Manager **2024 - 2026**

- Managed the chapter's LinkedIn presence, creating and scheduling posts to increase engagement and visibility.
- Recorded meeting minutes and coordinated chapter activities, supporting event planning and organizational operations.

CoST Young Scientist Day - North Carolina A&T State University **Greensboro, NC-USA**
Lead instructor, Workshop on Geospatial Sciences **2025**

- Led a Geospatial Sciences workshop, mentoring high school students on career pathways.
- Demonstrated drones, GIS software, and 3D DTM technologies in hands-on sessions.

BIOMES Seminar Committee – Yale University **New Haven, CT-USA**
Student Representative **2022 - 2023**

- Worked to invite speakers to speak at weekly Seminars and handle their logistics.
- Helped to facilitate students' engagement with speakers.

Good Lead Foundation **Nsuaem-Tarkwa, Ghana**
Co-Founder/Executive Director **2019 – Present**

- Leads the organization to initiate and implement projects.
- Help raise funds to support educational activities within our community.
- Organize seasonal internal capacity-building workshops for members of the Foundation.